Update
Mason 2YP

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1 Empirical demand

1.1 Notation

Symbol	Definition
$i \in \mathcal{I}$, $I = \lvert\mathcal{I}\rvert$	Household and number of retained households.
$t \in \mathcal{T}_i$	Retained household purchase week, ending Saturday.
$\mathcal{H}_i$	Observed purchase weeks used to construct history, including weeks subsequently excluded from the likelihood.
$\mathcal{F}$, Q_{ift}, w_{if}, H_i	Unaggregated flavor labels, quantity of detailed flavor f, household full-sample flavor quantity share, and household flavor HHI.
$j, k \in \mathcal{J}$	Flavor categories; $\mathcal{J} = \{\text{other}, \text{berry}, \text{plain}\}$.
0	Outside alternative, included in the reported fit.
Q_{ijt}, Q_{it}	Quantity purchased of flavor j; total inside quantity $Q_{it} = \sum_j Q_{ijt}$.
x_{ijt}	Inside quantity share of flavor j, Q_{ijt}/Q_{it}.
S_{ijt}, N_{ijt}	Cumulative own-flavor share stock and immediately preceding calendar week's other-flavor share.
λ, X_{ijt}	Fixed history weight and combined history index; $\lambda = 0.7$.
r, u	Store and UPC indices.
r_{it}, $\mathcal{U}_{rjt}$	Exposure store and observed, distinct price-covered UPCs for flavor j at store r in week t.
p_{rut}, p_{ijt}	Merged unit-price quote and exposure-store flavor price.
$\mathcal{A}_{it}$	Available inside flavors with finite positive exposure-store prices.
m, m_{it}	Consumer market index and household i's market in week t.
$\mathcal{Q}_{mt}$, n_{mt}, P_{mt}	Distinct store–week–UPC prices within consumer DMA m; their number and sum. Prices pool all flavors.
z_{it}	Same-week mean price outside consumer DMA m_{it}.
π_j, $\boldsymbol{\eta}_j$	First-stage instrument slope and coefficients on the three history indices.
τ_{jt}, κ_{jm}	Flavor-specific week and consumer-DMA fixed effects.
v_{ijt}, $\widehat{v}_{ijt}$	First-stage price effect and estimated residual.
$\mathcal{D}_j$, G_j	Flavor j first-stage household–weeks and number of distinct households therein.
b_0	Common inside intercept relative to the outside alternative.
β_j	Flavor intercept relative to other; $\beta_{\text{other}} = 0$ in the reported fit.
α_i, μ_α, σ_α	Positive price sensitivity and mean/SD of its logarithm. Price enters utility as $-\alpha_i p$.
γ_i, μ_γ, σ_γ	Household history coefficient, common across flavors, and its population mean/SD.
ρ	Common coefficient on the price-control residual.
ε_{ijt}, V_{ijt}, U_{ijt}	Unit-scale type-I extreme-value shock, deterministic utility, and total utility.
$Z_{i\alpha}$, $Z_{i\gamma}$	Independent standard normal household type components, fixed over household history.
$\mathcal{B}_{it}(C)$	Available alternatives, with outside added only for $C > 0$.
P_{ijt}	Conditional choice probability for given household type.
$\boldsymbol{\theta}$, Θ	Demand parameter vector and optimization bounds.
f, ϕ	Household-type density and standard normal density.
L_i, L_{ir}, ℓ_{ir}	Integrated household likelihood, draw-specific likelihood, and its logarithm; multinomial constants omitted.
R, $\boldsymbol{z}_r$	Number of integration draws and fixed antithetic standard normal draw. Draw index r is separate from the store index.
$\widehat{L}_i$, $\widehat{\ell}$, w_{ir}	Simulated likelihood, total simulated log likelihood, and household posterior draw weight.

Symbol	Definition
$\boldsymbol{g}_{ir}$	Draw-specific household likelihood score.
$\widehat{H}$, $\widehat{\Sigma}$, $\widehat{\mathrm{se}}$	Hessian of the mean negative log likelihood, estimated parameter covariance, and standard error.
g_i, F_j, $\widehat{\mathrm{Var}}(\widehat{\pi}_j)$	Household instrument score, first-stage one-instrument Wald statistic, and clustered variance estimate.

1.2 Price equation and control residuals

$$n_{mt} = |\mathcal{Q}_{mt}|, \tag{1}$$

$$P_{mt} = \sum_{q \in \mathcal{Q}_{mt}} p_q, \tag{2}$$

$$z_{it} = \frac{\sum_{m \neq m_{it}} P_{mt}}{\sum_{m \neq m_{it}} n_{mt}}, \qquad \sum_{m \neq m_{it}} n_{mt} > 0, \tag{3}$$

$$p_{ijt} = \pi_j z_{it} + \sum_{k \in \mathcal{J}} \eta_{jk} X_{ikt} + \tau_{jt} + \kappa_{j,m_{it}} + v_{ijt}, \qquad (i,t) \in \mathcal{D}_j. \tag{4}$$

The same z_{it} enters each flavor equation. Each household–week has equal first-stage weight.

Table 2: Price first stages

Flavor	$\widehat{\pi}_j$	Cluster SE	F_j	Partial R^2	Observations	Households
Other	-34.6795	0.9662	1,288.26	0.0213	252,033	14,965
Berry	-33.1049	1.3767	578.20	0.0192	94,753	11,434
Plain	-28.5850	3.0847	85.87	0.0076	41,835	7,510

1.3 Utility, heterogeneity, and choice probabilities

$$U_{ijt} = V_{ijt} + \varepsilon_{ijt}, \tag{5}$$

$$V_{ijt} = b_0 + \beta_j - \alpha_i p_{ijt} + \gamma_i X_{ijt} + \rho \widehat{v}_{ijt}, \tag{6}$$

$$\beta_{\text{other}} = 0, \tag{7}$$

$$\log \alpha_i = \mu_\alpha + \sigma_\alpha Z_{i\alpha}, \qquad \alpha_i > 0, \tag{8}$$

$$\gamma_i = \mu_\gamma + \sigma_\gamma Z_{i\gamma}, \tag{9}$$

$$\begin{pmatrix} Z_{i\alpha} \\ Z_{i\gamma} \end{pmatrix} \sim N(\mathbf{0}, I_2), \tag{10}$$

$$P_{ijt}(\alpha_i, \gamma_i) = \frac{\exp(V_{ijt})}{1 + \sum_{k \in \mathcal{A}_{it}} \exp(V_{ikt})}, \qquad j \in \mathcal{A}_{it}, \tag{11}$$

$$\boldsymbol{\theta} = (b_0, \beta_{\text{berry}}, \beta_{\text{plain}}, \mu_\alpha, \mu_\gamma, \sigma_\alpha, \sigma_\gamma, \rho)^\top. \tag{12}$$

$$\mathbb{E}[\alpha_i] = \exp(\mu_\alpha + \sigma_\alpha^2/2), \tag{13}$$

$$\mathrm{Var}(\alpha_i) = [\exp(\sigma_\alpha^2) - 1]\exp(2\mu_\alpha + \sigma_\alpha^2), \tag{14}$$

$$\mathbb{E}[\gamma_i] = \mu_\gamma, \tag{15}$$

$$\mathrm{Var}(\gamma_i) = \sigma_\gamma^2, \tag{16}$$

$$f(\alpha, \gamma) = \frac{1}{\alpha\sigma_\alpha}\phi\left(\frac{\log\alpha - \mu_\alpha}{\sigma_\alpha}\right)\frac{1}{\sigma_\gamma}\phi\left(\frac{\gamma - \mu_\gamma}{\sigma_\gamma}\right). \tag{17}$$

Capacity and outside alternative

$$Q_{i0t} = \max\{C - Q_{it}, 0\}, \qquad C > 0, \tag{18}$$

$$V_{i0t} = 0, \qquad V_{ijt} = b_0 + \beta_j - \alpha_i p_{ijt} + \gamma_i X_{ijt} + \rho\widehat{v}_{ijt}, \tag{19}$$

$$\mathcal{B}_{it}(C) = \begin{cases} \mathcal{A}_{it}, & C = 0, \\ \mathcal{A}_{it} \cup \{0\}, & C > 0, \end{cases} \tag{20}$$

$$P_{ijt} = \frac{\exp(V_{ijt})}{\sum_{k\in\mathcal{B}_{it}(C)}\exp(V_{ikt})}. \tag{21}$$

Reported results: $C = 7$, $Q_{i0t} = \max\{7 - Q_{it}, 0\}$, and b_0 is estimated; $\mathcal{B}_{it}(7) = \mathcal{A}_{it} \cup \{0\}$. Capacity imputes outside quantity on observed purchase weeks; it is not an upper bound on inside purchases. Multinomial combinatorial constants are omitted.

1.4 Household likelihood and simulation

$$L_i(\boldsymbol{\theta}) = \int_0^\infty \int_{-\infty}^\infty \left[\prod_{t\in\mathcal{T}_i}\prod_{j\in\mathcal{B}_{it}(7)} P_{ijt}(\alpha,\gamma)^{Q_{ijt}}\right] f(\alpha,\gamma)\,\mathrm{d}\gamma\,\mathrm{d}\alpha, \tag{22}$$

$$\boldsymbol{z}_r = (z_{r\alpha}, z_{r\gamma})^\top, \quad \boldsymbol{z}_{r+R/2} = -\boldsymbol{z}_r, \quad \boldsymbol{z}_r \overset{\text{iid}}{\sim} N(\mathbf{0}, I_2), \quad r \le R/2, \tag{23}$$

$$\alpha_r = \exp(\mu_\alpha + \sigma_\alpha z_{r\alpha}), \tag{24}$$

$$\gamma_r = \mu_\gamma + \sigma_\gamma z_{r\gamma}, \tag{25}$$

$$\ell_{ir} = \sum_{t\in\mathcal{T}_i}\sum_{j\in\mathcal{B}_{it}(7)} Q_{ijt}\log P_{ijt}(\alpha_r, \gamma_r), \tag{26}$$

$$L_{ir} = e^{\ell_{ir}}, \tag{27}$$

$$\widehat{L}_i = \frac{1}{R}\sum_{r=1}^R e^{\ell_{ir}}, \tag{28}$$

$$\log\widehat{L}_i = \mathrm{LSE}_{r=1}^R(\ell_{ir}) - \log R, \qquad \mathrm{LSE}(a_1,\ldots,a_R) = \log\sum_r e^{a_r}, \tag{29}$$

$$\widehat{\ell}(\boldsymbol{\theta}) = \sum_{i=1}^I \log\widehat{L}_i(\boldsymbol{\theta}), \tag{30}$$

$$\widehat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}\in\Theta}\widehat{\ell}(\boldsymbol{\theta}), \tag{31}$$

$$\Theta = [-20, 20]^3 \times [-6, 3] \times [-20, 20] \times [0, 3]^2 \times [-20, 20]. \tag{32}$$

The optimizer minimizes $-\widehat{\ell}/I$.

1.5 Standard errors

The estimation sample is restricted to purchase records with household size one before weekly quantities and history indices are constructed. Store price menus and retail instruments use all matched purchase records.

Let $m(\boldsymbol{\theta}) = -\widehat{\ell}(\boldsymbol{\theta})/I$ denote the mean household negative simulated log likelihood. Define the observed information and covariance estimator as

$$\widehat{H} = \left.\frac{\partial^2 m(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}^\top}\right|_{\boldsymbol{\theta}=\widehat{\boldsymbol{\theta}}}, \qquad \widehat{\Sigma} = \frac{1}{I}\widehat{H}^{-1}. \tag{33}$$

The Hessian is computed by central differences of the analytic gradient, using the same fixed integration draws as estimation. Standard errors and normal 95% confidence intervals are

$$\widehat{\mathrm{se}}(\widehat{\theta}_k) = \sqrt{\widehat{\Sigma}_{kk}}, \qquad \mathrm{CI}_{k,.95} = \widehat{\theta}_k \pm 1.96\, \widehat{\mathrm{se}}(\widehat{\theta}_k). \tag{34}$$

These model-based standard errors condition on the estimated price-control residuals, retail instruments, and integration draws; they omit first-stage estimation uncertainty. Standard errors are withheld for a nonconverged fit, parameters at or near bounds, or nonpositive or ill-conditioned information.

1.6 Unaggregated flavor concentration

$$w_{if} = \frac{\sum_{t\in\mathcal{H}_i} Q_{ift}}{\sum_{g\in\mathcal{F}} \sum_{t\in\mathcal{H}_i} Q_{igt}}, \tag{35}$$

$$\sum_{f\in\mathcal{F}} w_{if} = 1, \tag{36}$$

$$H_i = \sum_{f\in\mathcal{F}} w_{if}^2. \tag{37}$$

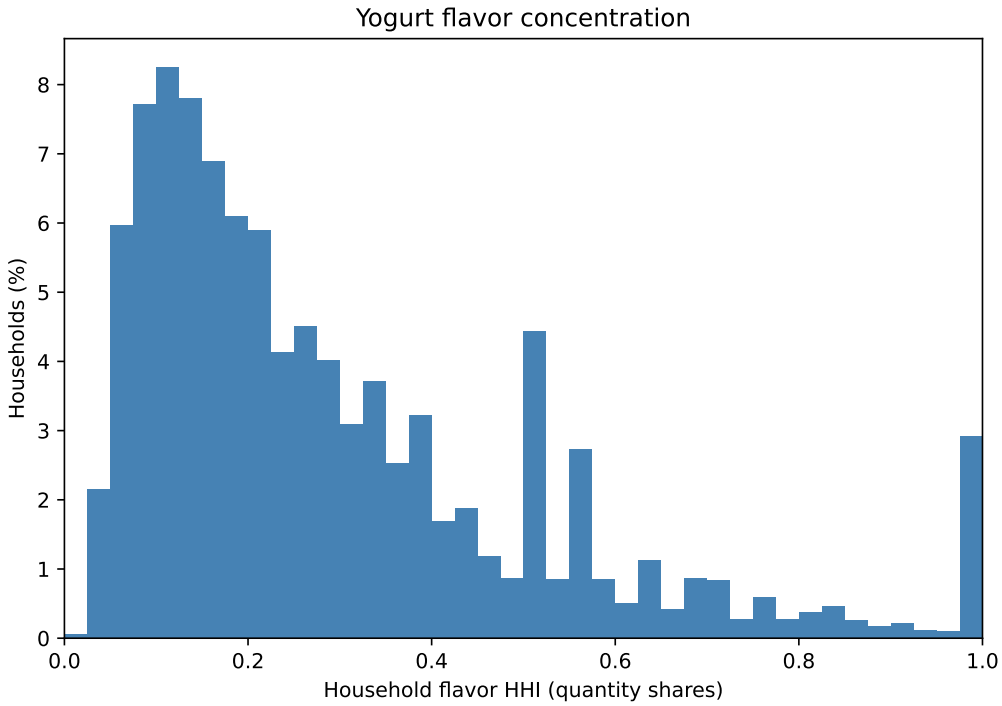


Figure 1: Household flavor HHI; unaggregated flavor labels, shares of quantity purchased.

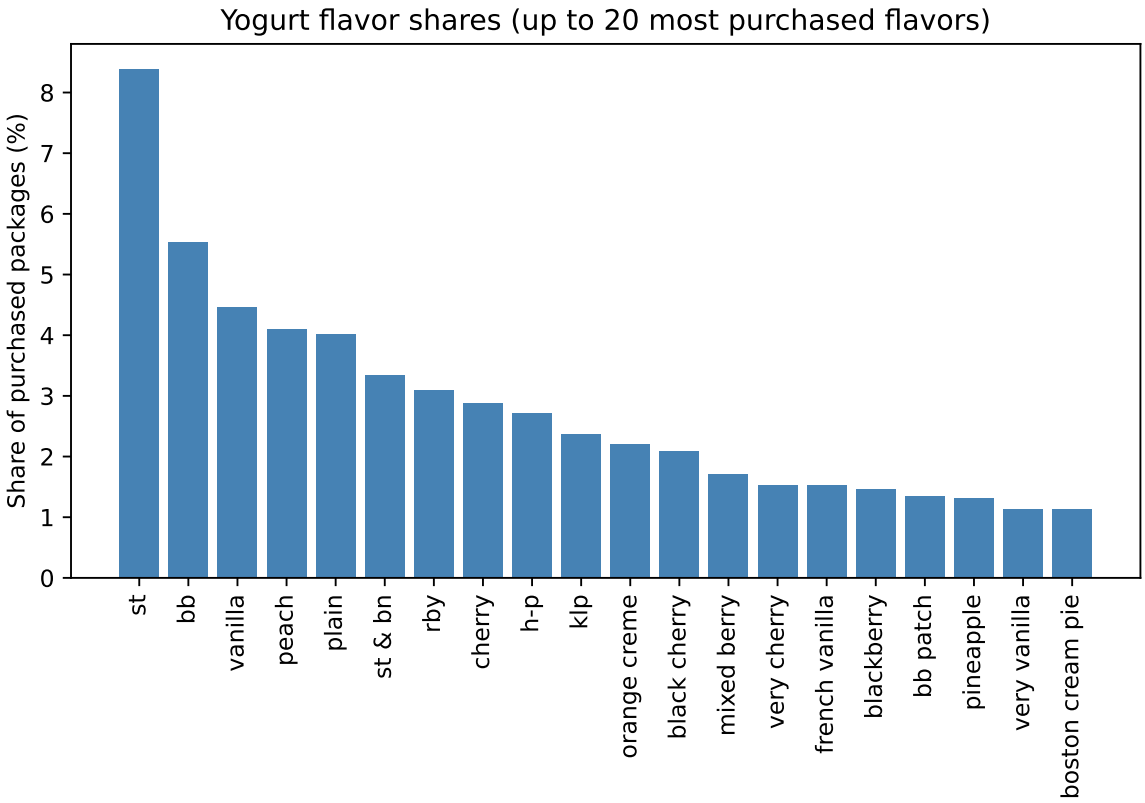


Figure 2: Shares of quantity purchased; 20 most purchased unaggregated flavor labels.

1.7 Summary statistics

The descriptive sample covers 2006–2015. Statistics are unweighted. Trips, UPCs, purchase lines, and quantity purchased in the matched yogurt column refer to yogurt only. The descriptive matched sample includes households of all sizes and precedes the single-person restriction used for demand estimation.

Table 3: Household purchase summaries

Statistic	Full panel	Matched yogurt
Households	148,045	70,153
Mean observed panel years	3.99	3.28
Mean trips per household	576.29	22.87
Mean trips per approximate observed week	2.54	0.12
Median trips per approximate observed week	2.15	0.08
Mean distinct UPCs per household	1,699.74	16.52
Mean purchase lines per household	5,082.26	52.74
Mean quantity purchased per household	7,036.50	115.45
One-person households (%)	20.39	18.40
Two-person households (%)	37.72	37.68
Hispanic origin (%)	6.33	6.13

Table 4: Yogurt purchase dynamics: detailed flavors and estimation categories

Statistic	Detailed flavors	Three categories
Households	70,153	70,153
Observed purchase household–weeks	1,462,897	1,462,897
Mean flavors/categories bought per household	12.89	1.97
Mean household flavor HHI	0.29	0.78
Mixed-flavor/category weeks (%)	40.08	18.07
Weeks with quantity purchased above seven (%)	25.17	25.17
Eligible adjacent single-flavor/category week pairs	148,604	255,708
Switch rate among eligible pairs (%)	55.04	13.58
Mean single-flavor/category spell length (weeks)	1.08	1.23
Median gap between purchase weeks (weeks)	3.00	3.00

An eligible transition requires purchases in consecutive calendar weeks and exactly one flavor (or category) in each week. Mixed baskets and gaps are excluded from the switch-rate denominator. Spells count consecutive calendar weeks with the same single flavor or category; a gap breaks the spell. HHI uses each household's full-history shares of quantity purchased.

Table 5: Quantity shares and transitions for the three estimation categories

	Quantity share (%)	Next category (row probabilities, %)		
Category		Other	Berry	Plain
Other	80.22	92.12	5.97	1.91
Berry	14.36	53.54	44.61	1.85
Plain	5.42	25.32	2.30	72.38

1.8 Results

Table 6: Estimation sample and settings; single-person households

Setting	Value
Years	2006–2015
Households	15,105
Household–weeks	281,695
Capacity setting	7
λ	0.7
Integration draws	32
Optimization starts	1
Household size	1
Converged	Yes
Parameters at bounds	0
Negative log likelihood	1,642,195.0310

Table 7: Demand estimates

Parameter	Estimate	Standard error	95% normal interval
b_0	1.3486	0.0066	[1.3356, 1.3616]
β_{berry}	-0.6504	0.0032	[-0.6567, -0.6440]
β_{plain}	-0.1040	0.0063	[-0.1164, -0.0916]
μ_α	-0.3620	0.0068	[-0.3754, -0.3486]
μ_γ	-0.0588	0.0002	[-0.0593, -0.0583]
σ_α	0.6654	0.0045	[0.6567, 0.6742]
σ_γ	0.0634	0.0003	[0.0629, 0.0639]
ρ	0.1675	0.0044	[0.1588, 0.1762]

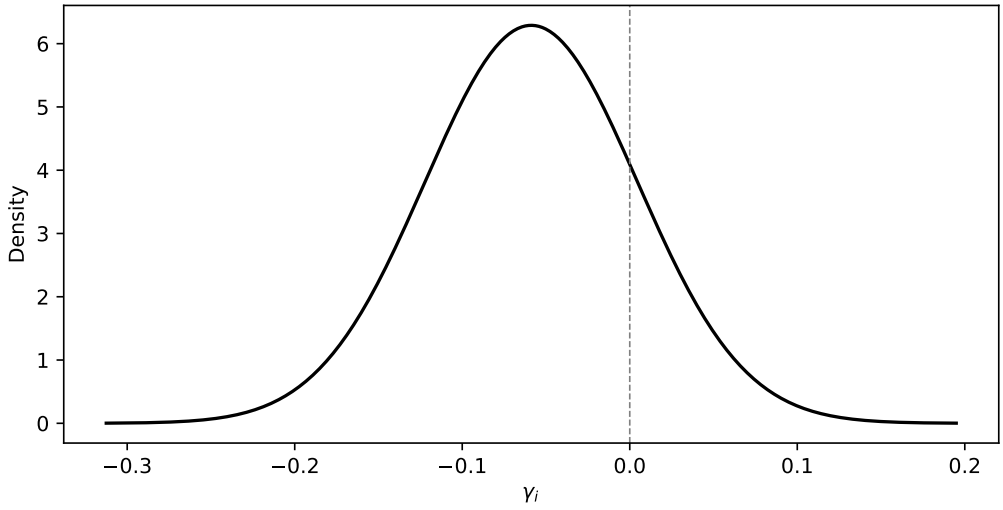


Figure 3: Fitted household history coefficient: $\gamma_i \sim N(-0.0588034,\, 0.0634273^2)$.

1.9 Discussion of Results

I now have 15,105 single-person households in my estimation sample.

Looking at the within household flavor concentration plot shows that most consumers consumer many flavors of yogurt. We do see a large grouping at the single flavor point, but they still only account for about 3% of households. Interestingly, there is another large cluster around the 50% point, which is good evidence of switching being for novelty rather than experimentation/learning. Strawberry yogurt is the most purchased flavor at about 8.5% of all sales. Notably, nine berry flavors populate the top 20 flavors, showing that it is a good grouping. I do think aggregating "other" into "other fruits", "dessert flavors", and "other" may be good once the three category specification is working as we want it to.

2 Supply: two-period pricing

2.1 Notation

Symbol	Definition
$t \in \{1,2\}$, j, 0	Period, inside product, and outside alternative.
$x_0 = h$, $x_1 = a$	Initial realized choice and realized first-period choice; each belongs to $\{j, 0\}$.
β, $\alpha > 0$, γ	Fixed, firm-observed type's inside intercept, positive price sensitivity, and symmetric switching coefficient. Type indices are suppressed.
c, $\delta \in (0,1)$	Marginal cost and discount factor.
$p_t(h)$	Price conditional on preceding realized choice h. Prices belong to $\mathbb{R}$.
V_{jt}^{raw}, V_{0t}^{raw}	Raw deterministic utilities of inside and outside alternatives.
$v_t(h)$	Inside-minus-outside deterministic utility.
$q(h)$	Signed history indicator: $q(j) = -1$, $q(0) = 1$.
ε_{jt}, ε_{0t}	Independent unit-scale type-I extreme-value shocks across alternatives and periods.
$s_t(h)$, $s_{0t}(h)$	Inside and outside choice probabilities.
$W_2(a)$	Optimized terminal profit conditional on $x_1 = a$.
ΔW	$W_2(j) - W_2(0)$.
$\Pi(p_1; h)$, $\Pi^*(h)$	Expected discounted profit at candidate first-period price and at its optimum.
s, r, u, v, B_2	Algebraic abbreviations only: $s = s_1^*(h)$, $r = 1 - s$, $u = s_2^*(j)$, $v = s_2^*(0)$, $B_2 = u + v$.
F_1, F_2	Implicit first- and second-period optimal-price equations after dividing the profit FOC by the share derivative.
b, k, $H(p)$	Generic logit intercept, effective cost, and sign-equivalent FOC function in the uniqueness calculation.
$p^{\text{stat}}(h)$, $s^{\text{stat}}(h)$	Separately optimized static price and its associated share ($\delta = 0$).
D_h, E_h	Numerators defined below for the first-period price and profit derivatives.
R_j, R_0, $R_{0\to j}$	Two state-matched price ratios and the ratio conditional on path $0 \to j$.
∂, $\mathrm{d}/\mathrm{d}\gamma$, prime, $*$	Partial derivative, total derivative at fixed discrete histories, total gamma derivative, and optimized quantity.

2.2 Utilities and shares

$$V_{jt}^{\text{raw}} = \beta - \alpha p_t + \gamma \mathbf{1}\{x_{t-1} = 0\}, \tag{38}$$

$$V_{0t}^{\text{raw}} = \gamma \mathbf{1}\{x_{t-1} = j\}, \tag{39}$$

$$\begin{aligned} v_t(h) &= V_{jt}^{\text{raw}} - V_{0t}^{\text{raw}} \\ &= \beta - \alpha p_t(h) + \gamma[\mathbf{1}\{h = 0\} - \mathbf{1}\{h = j\}] \\ &= \beta - \alpha p_t(h) + q(h)\gamma, \end{aligned} \tag{40}$$

$$s_t(h) = \frac{e^{v_t(h)}}{1 + e^{v_t(h)}}, \qquad s_{0t}(h) = 1 - s_t(h). \tag{41}$$

$$\frac{\partial s_t}{\partial p_t} = -\alpha s_t(1 - s_t), \tag{42}$$

$$\left.\frac{\partial s_t}{\partial \gamma}\right|_{p_t,h} = q(h) s_t(1 - s_t), \tag{43}$$

2.3 Terminal-period optimization

For each $a \in \{j, 0\}$:

$$W_2(a) = \max_{p_2(a) \in \mathbb{R}} s_2(a)[p_2(a) - c], \tag{44}$$

$$\begin{aligned} 0 &= \frac{\partial}{\partial p_2}\{s_2(p_2 - c)\} \\ &= s_2 + (p_2 - c)\frac{\partial s_2}{\partial p_2} \\ &= s_2 - \alpha s_2(1 - s_2)(p_2 - c), \end{aligned} \tag{45}$$

$$1 = \alpha(1 - s_2^*)(p_2^* - c), \tag{46}$$

$$p_2^*(a) = c + \frac{1}{\alpha[1 - s_2^*(a)]}, \tag{47}$$

$$W_2(a) = \frac{s_2^*(a)}{\alpha[1 - s_2^*(a)]}. \tag{48}$$

2.4 First-period optimization

$$\Delta W = W_2(j) - W_2(0), \tag{49}$$

$$\begin{aligned} \Pi(p_1; h) &= s_1(h)(p_1 - c) + \delta\{s_1(h) W_2(j) + [1 - s_1(h)] W_2(0)\} \\ &= \delta W_2(0) + s_1(h)[p_1 - c + \delta \Delta W], \end{aligned} \tag{50}$$

$$\begin{aligned} \partial_{p_1}\Pi &= s_1 + \partial_{p_1} s_1 [p_1 - c + \delta \Delta W] \\ &= s_1 - \alpha s_1(1 - s_1)[p_1 - c + \delta \Delta W], \end{aligned} \tag{51}$$

$$0 = 1 - \alpha(1 - s_1^*)[p_1^* - c + \delta \Delta W], \tag{52}$$

$$p_1^*(h) = c + \frac{1}{\alpha[1 - s_1^*(h)]} - \delta\Delta W, \tag{53}$$

$$\Pi^*(h) = \delta W_2(0) + \frac{s_1^*(h)}{\alpha[1 - s_1^*(h)]}. \tag{54}$$

The two terminal value functions are constant with respect to p_1 conditional on the realized terminal state.

2.5 Terminal-price and terminal-value derivatives

For fixed a, write $s_2 = s_2^*(a)$ and $q = q(a)$ within this subsection.

$$F_2(p_2, \gamma; a) = p_2 - c - \frac{1}{\alpha(1 - s_2)}, \tag{55}$$

$$\partial_{p_2} F_2 = 1 - \frac{1}{\alpha}\frac{\partial_{p_2} s_2}{(1 - s_2)^2} = 1 + \frac{s_2}{1 - s_2} = \frac{1}{1 - s_2}, \tag{56}$$

$$\partial_\gamma F_2 = -\frac{1}{\alpha}\frac{q s_2(1 - s_2)}{(1 - s_2)^2} = -\frac{q s_2}{\alpha(1 - s_2)}, \tag{57}$$

$$\frac{\mathrm{d}p_2^*(a)}{\mathrm{d}\gamma} = -\frac{\partial_\gamma F_2}{\partial_{p_2} F_2} = \frac{q(a) s_2^*(a)}{\alpha}, \tag{58}$$

$$\begin{aligned} \frac{\mathrm{d}s_2^*(a)}{\mathrm{d}\gamma} &= s_2(1 - s_2)\left[q(a) - \alpha\frac{\mathrm{d}p_2^*(a)}{\mathrm{d}\gamma}\right] \\ &= q(a) s_2(1 - s_2)^2, \end{aligned} \tag{59}$$

$$\begin{aligned} \frac{\mathrm{d}W_2(a)}{\mathrm{d}\gamma} &= \partial_\gamma\{s_2(p_2 - c)\}|_{p_2 = p_2^*} \\ &= q(a) s_2(1 - s_2)(p_2^* - c) = \frac{q(a) s_2}{\alpha}. \end{aligned} \tag{60}$$

Restore $u = s_2^*(j)$ and $v = s_2^*(0)$:

$$\frac{\mathrm{d}p_2^*(j)}{\mathrm{d}\gamma} = -\frac{u}{\alpha}, \tag{61}$$

$$\frac{\mathrm{d}p_2^*(0)}{\mathrm{d}\gamma} = \frac{v}{\alpha}, \tag{62}$$

$$W_2'(j) = -\frac{u}{\alpha}, \tag{63}$$

$$W_2'(0) = \frac{v}{\alpha}, \tag{64}$$

$$\Delta W' = -\frac{u + v}{\alpha} = -\frac{B_2}{\alpha}, \tag{65}$$

$$u' = -u(1 - u)^2, \tag{66}$$

$$v' = v(1 - v)^2, \tag{67}$$

$$B_2' = -u(1 - u)^2 + v(1 - v)^2. \tag{68}$$

2.6 First-period price and share derivatives

Let $q = q(h)$, $s = s_1^*(h)$, $r = 1 - s$.

$$F_1(p_1, \gamma; h) = p_1 - c - \frac{1}{\alpha(1 - s_1)} + \delta\Delta W, \tag{69}$$

$$\partial_{p_1} F_1 = 1 - \frac{1}{\alpha}\frac{-\alpha s_1(1 - s_1)}{(1 - s_1)^2} = \frac{1}{r}, \tag{70}$$

$$\partial_\gamma F_1 = -\frac{qs}{\alpha r} + \delta\Delta W' = -\frac{qs}{\alpha r} - \frac{\delta B_2}{\alpha}, \tag{71}$$

$$p_1^{*\prime}(h) = -\frac{\partial_\gamma F_1}{\partial_{p_1} F_1} = \frac{qs + \delta r B_2}{\alpha}, \tag{72}$$

$$\begin{aligned} s' &= sr(q - \alpha p_1^{*\prime}) \\ &= sr[q - qs - \delta r B_2] = sr^2(q - \delta B_2), \end{aligned} \tag{73}$$

$$r' = -s'. \tag{74}$$

$$\begin{aligned} p_1^{*\prime\prime} &= \frac{qs' + \delta r' B_2 + \delta r B_2'}{\alpha} \\ &= \frac{(q - \delta B_2)s' + \delta r B_2'}{\alpha} \\ &= \frac{sr^2(q - \delta B_2)^2 + \delta r B_2'}{\alpha}. \end{aligned} \tag{75}$$

2.7 Profit derivative

$$\begin{aligned} \frac{\mathrm{d}\Pi^*(h)}{\mathrm{d}\gamma} &= \partial_\gamma \Pi(p_1; h)|_{p_1 = p_1^*} \\ &= qsr[p_1^* - c + \delta\Delta W] + \delta\{sW_2'(j) + rW_2'(0)\} \\ &= qsr\frac{1}{\alpha r} + \delta\left(-\frac{su}{\alpha} + \frac{rv}{\alpha}\right) \\ &= \frac{qs - \delta su + \delta rv}{\alpha}. \end{aligned} \tag{76}$$

2.8 Conditional derivative signs

$$D_h = q(h)s + \delta r(u + v), \tag{77}$$

$$p_1^{*\prime}(h) = D_h/\alpha, \tag{78}$$

$$E_h = q(h)s - \delta su + \delta rv, \tag{79}$$

$$\Pi^{*\prime}(h) = E_h/\alpha. \tag{80}$$

$$h = 0: \quad D_0 = s + \delta r(u + v) > 0, \tag{81}$$

$$E_0 = s(1 - \delta u) + \delta rv > 0, \tag{82}$$

$$h = j: \quad D_j = -s + \delta r(u+v), \tag{83}$$

$$E_j = -s(1+\delta u) + \delta r v, \tag{84}$$

$$p_1^{*\prime}(j) > 0 \iff \delta r(u+v) > s, \tag{85}$$

$$\Pi^{*\prime}(j) > 0 \iff \delta r v > s(1+\delta u). \tag{86}$$

Equality gives a zero derivative; reversing each inequality reverses the corresponding sign.

2.9 Price ratios and their derivatives

$$R_j = \frac{p_1^*(j)}{p_2^*(j)}, \tag{87}$$

$$R_0 = \frac{p_1^*(0)}{p_2^*(0)}, \tag{88}$$

$$R_{0\to j} = \frac{p_1^*(0)}{p_2^*(j)}, \tag{89}$$

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}\gamma}\left(\frac{p_1^*}{p_2^*}\right) &= \frac{1}{p_2^*}\frac{\mathrm{d}p_1^*}{\mathrm{d}\gamma} - \frac{p_1^*}{(p_2^*)^2}\frac{\mathrm{d}p_2^*}{\mathrm{d}\gamma} \\ &= \frac{p_2^* p_1^{*\prime} - p_1^* p_2^{*\prime}}{(p_2^*)^2}, \end{aligned} \tag{90}$$

$$R_j' = \frac{p_2^*(j)p_1^{*\prime}(j) + p_1^*(j)u/\alpha}{[p_2^*(j)]^2}, \tag{91}$$

$$R_0' = \frac{p_2^*(0)p_1^{*\prime}(0) - p_1^*(0)v/\alpha}{[p_2^*(0)]^2}, \tag{92}$$

$$R_{0\to j}' = \frac{p_2^*(j)p_1^{*\prime}(0) + p_1^*(0)u/\alpha}{[p_2^*(j)]^2}. \tag{93}$$

Each ratio requires a nonzero denominator. Each derivative holds its stated discrete histories fixed.

2.10 Numerical specification

$\beta = 2, \qquad \alpha = 1, \qquad \delta = 0.92, \qquad c = 0, \qquad \gamma \in [-3, 3]$ (601 grid points per initial state).

Table 9: Selected supply solutions

h	γ	$p_1^*(h)$	$p_2^*(j)$	$p_2^*(0)$	$\Pi^*(h)$	$\mathrm{d}\Pi^*(h)/\mathrm{d}\gamma$
j	-3	3.395	3.926	1.120	5.087	-1.387
j	-1	2.236	2.557	1.567	2.669	-0.959
j	0	2.000	2.000	2.000	1.920	-0.500
j	1	2.209	1.567	2.557	1.731	0.125
j	3	3.592	1.120	3.926	2.702	0.668
0	-3	-0.780	3.926	1.120	0.913	0.195
0	-1	1.045	2.557	1.567	1.478	0.385
0	0	2.000	2.000	2.000	1.920	0.500
0	1	2.956	1.567	2.557	2.478	0.615
0	3	4.801	1.120	3.926	3.912	0.804

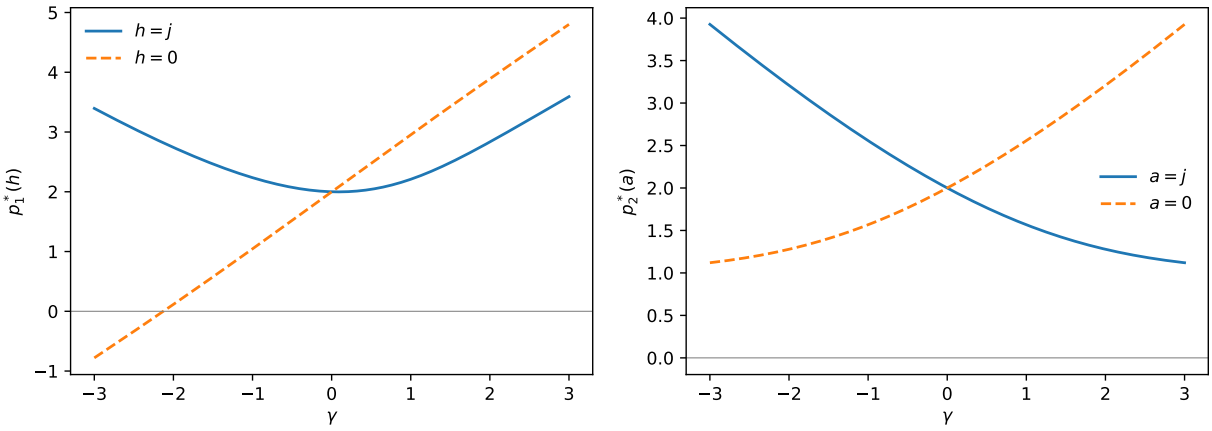


Figure 4: $p_1^*(h)$ and $p_2^*(a)$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

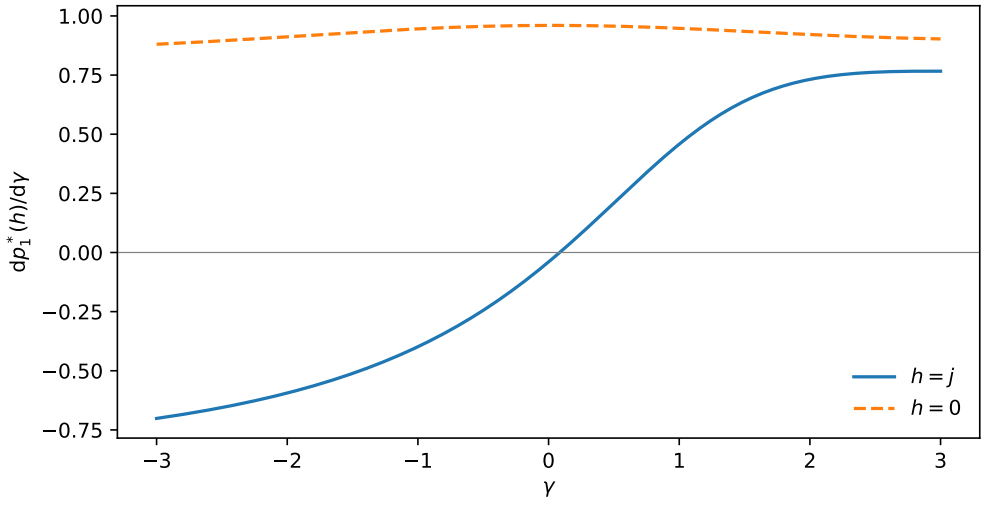


Figure 5: $\mathrm{d}p_1^*(h)/\mathrm{d}\gamma$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

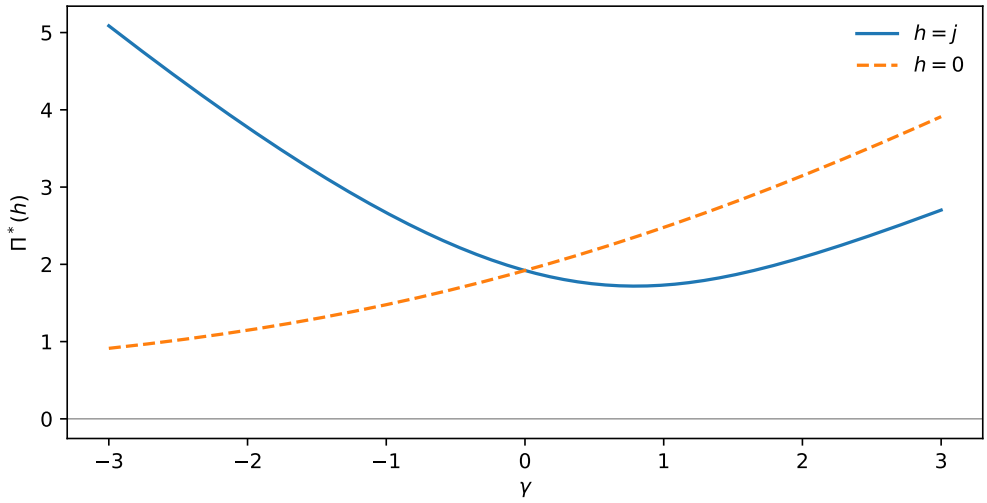


Figure 6: $\Pi^*(h)$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

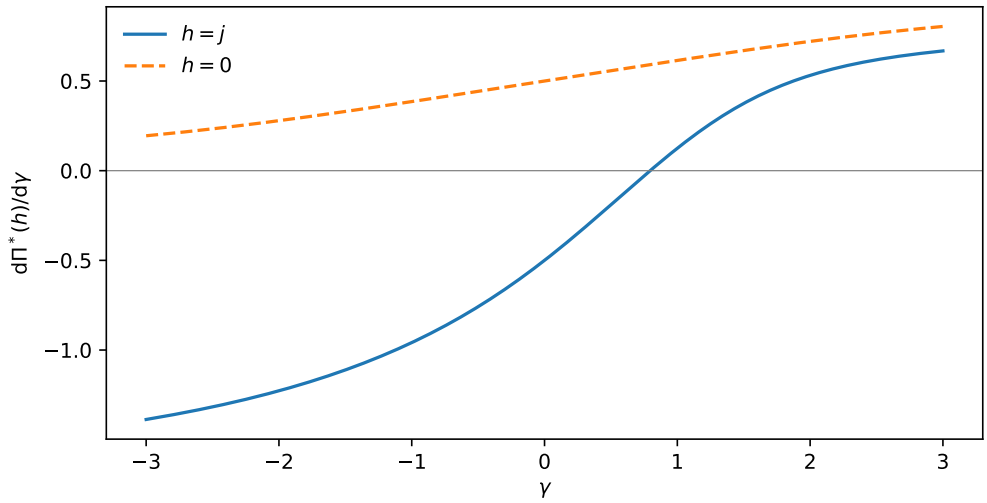


Figure 7: $\mathrm{d}\Pi^*(h)/\mathrm{d}\gamma$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

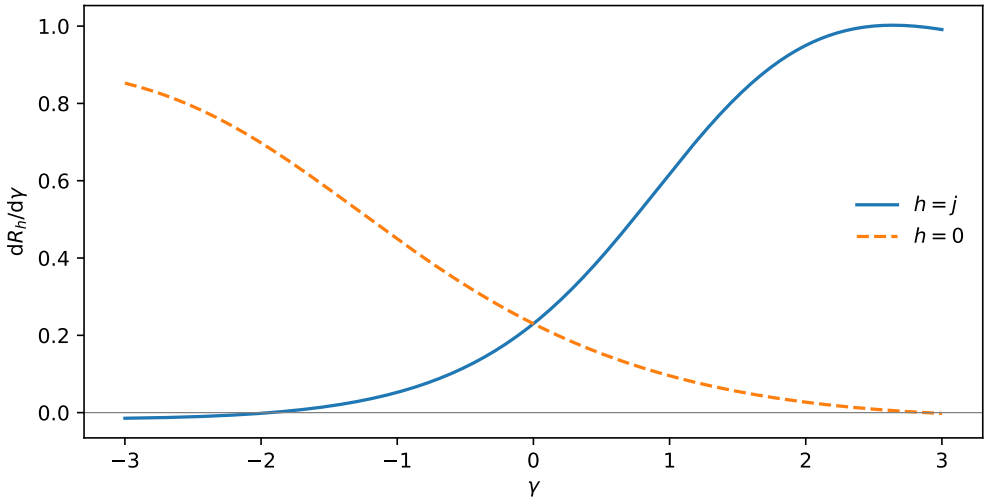


Figure 8: $\mathrm{d}R_j/\mathrm{d}\gamma$ and $\mathrm{d}R_0/\mathrm{d}\gamma$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

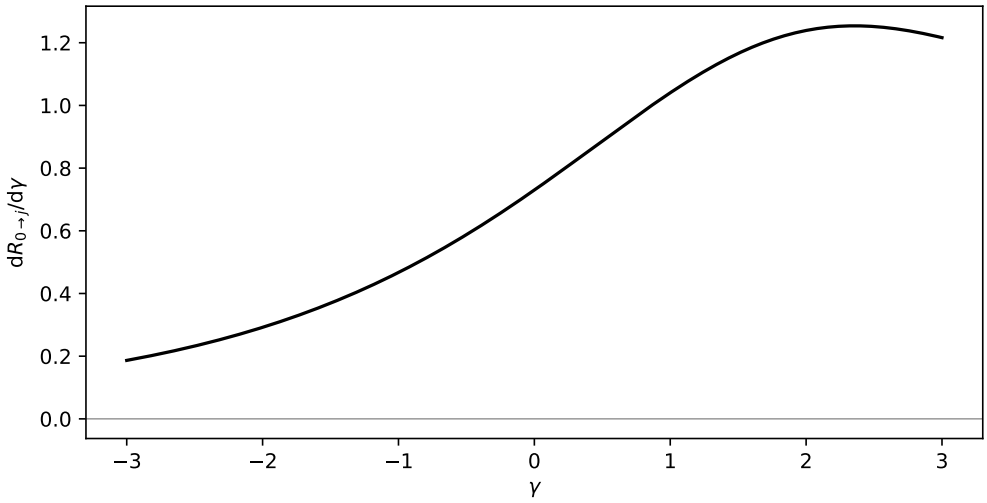


Figure 9: $\mathrm{d}R_{0\to j}/\mathrm{d}\gamma$; $\beta = 2$, $\alpha = 1$, $\delta = 0.92$, $c = 0$.

2.11 Discussion of Supply

The optimal pricing conditions show that the period price one is convex for consumers with initial state $h_0 = j$, implying higher prices for individuals at the poles of the γ distribution. For $h_0 = 0$, the firm will raise its price as γ increases. This tracks with what I would expect to see. As a consumer is more likely to switch, they will be less impacted by a higher price. Period two prices move in opposite directions. Those who bought $a = j$ in period one see a lower price as the level of γ increases, attempting to hold them, while $a = 0$ last period sees higher prices as the level of γ increases. From this we can see that switchers are charged higher prices at higher levels of switching preference in both periods. This points towards an idea of trying to charge as much as possible before the consumer switches away. Similarly, as consumer's prefer to show persistence in consumption, period two prices are very high. Optimal prices are approximately four times higher for stayers than switchers in period two. We perfectly see the induce then harvest mechanism here. At $\gamma = 3$ the firm is even willing to subsidize the consumer to switch in order to charge a much higher price in the next period.

From the derivative of optimal period one price w.r.t. γ, initial state $h_0 = 0$ we have a relatively flat effect, seeing slight concavity. However, the curve for $h_0 = j$ shows that the effect of γ is quite variable depending on consumer type.

Optimal profit shows that for $h_0 = j$, higher levels of γ reduce profits until $\gamma = 1$ when the curve turns back up, implying there is a level of switching preference which becomes profitable to target. For $h_0 = 0$, higher values of γ are more profitable since if they can get the consumer to switch in period one, they can harvest in period two. However, this is a bit difficult for me to interpret since if they are a high type switcher, they would likely switch in the path $0 \to j \to 0$, reducing firm profits. Something to continue to analyze on my part.

The profit derivative w.r.t. γ is remarkably similar to the optimal first period derivative for $h_0 = j$, though for $h_0 = 0$ there is much more curvature, showing that stayers are less profitable since they will likely remain with the outside option.

The pricing ratio derivatives show three cases: $j \to j = R_j,\ 0 \to 0 = R_0,\ 0 \to j = R_{0 \to j}$. In the inside only case, period one prices dominate with increases in γ, seeing the value of the ratio grow with γ. For the outside only, we see the opposite. This is again due to induce and harvest. The firm will charge a lower period one price then charge a higher period two price for high switchers, leading to the downward slope. Finally, in the switching case, the curve looks remarkably similar to the inside only case, though it starts at a higher point. Induce and harvest is well demonstrated by this figure, showing higher period two prices for those likely to stay with j if they switch from initial state $0 \to j$ in period one. As the consumer becomes more likely to switch, the firm will attempt to charge as much as possible in period one.

These results demonstrate the firm's objective of targeting to first induce customers to their product via discounts, then ratcheting up prices in subsequent periods to reap profit from new consumers. A future task would be to put the estimated distribution from the empirical exercise into this this simulation to see how the firm would actually price and then compare to actual pricing.